

MAGNETIC FLUX & FARADAY'S LAW

Flux *Wb*

$$\Phi = \iint_S \mathbf{B} \cdot d\mathbf{s}$$

Faraday's law *N turns; N = 1 for single loop*

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} \quad (\text{V})$$

Lenz's law: $\mathbf{B}$ increasing $\Rightarrow \mathbf{B}_{\text{ind}}$ **opposes** $\mathbf{B}$; $\mathbf{B}$ decreasing $\Rightarrow \mathbf{B}_{\text{ind}}$ same dir.

TRANSFORMER & MOTIONAL EMF

Transformer EMF *stationary loop, B changes*

$$V_{\text{emf}}^{\text{tr}} = -N \iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

Motional EMF *moving loop, static B*

$$V_{\text{emf}}^{\text{m}} = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

Total (general) *both loop moves and B varies*

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}}$$

If $\mathbf{u} \parallel d\mathbf{l}$ or $\mathbf{B} \parallel d\mathbf{l}$: that segment contributes 0 to $V_{\text{emf}}^{\text{m}}$.

AC GENERATOR

Loop area A , N turns, angular vel. ω , initial angle C_0 , field B_0 :

$$\Phi = AB_0 \cos(\omega t + C_0), \quad V_{\text{emf}} = N A \omega B_0 \sin(\omega t + C_0)$$

Peak $V_0 = N A \omega B_0$. To increase: $\uparrow N, A, \omega, B_0$.

IDEAL TRANSFORMER

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}, \quad \frac{I_1}{I_2} = \frac{N_2}{N_1}$$
$$Z_{\text{in}} = \left(\frac{N_1}{N_2}\right)^2 Z_L, \quad R_{\text{in}} = \left(\frac{N_1}{N_2}\right)^2 R_L$$

Inductor in changing $\mathbf{B}$: $I = V_{\text{emf}}^{\text{tr}}/(R + R_i) \approx V_{\text{emf}}^{\text{tr}}/R$ when $R_i \ll R$.

MAXWELL'S EQUATIONS

Law	Differential	Integral
Gauss $\mathbf{E}$	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint \mathbf{D} \cdot d\mathbf{s} = Q$
Gauss $\mathbf{B}$	$\nabla \cdot \mathbf{B} = 0$	$\oint \mathbf{B} \cdot d\mathbf{s} = 0$
Faraday	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}$
Ampère	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint \mathbf{H} \cdot d\mathbf{l} = I_c + I_d$
$\mathbf{D} = \epsilon \mathbf{E}; \mathbf{B} = \mu \mathbf{H}; \mathbf{J} = \sigma \mathbf{E}; I_d = \iint \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}; I_c = \iint \mathbf{J} \cdot d\mathbf{s}.$		
Phasor: $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}; \nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}; \epsilon_c = \epsilon' - j\sigma/\omega.$		

DISPLACEMENT CURRENT

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} = \epsilon \frac{\partial \mathbf{E}}{\partial t} \quad (\text{A/m}^2), \quad I_d = \iint_S \mathbf{J}_d \cdot d\mathbf{s} \quad (\text{A})$$

Phasor: $\tilde{\mathbf{J}}_d = j\omega\epsilon\tilde{\mathbf{E}}$. Makes total current $I = I_c + I_d$ continuous.

CHARGE CONTINUITY & RELAXATION

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \implies \oint_S \mathbf{J} \cdot d\mathbf{s} = 0 \quad (\text{KCL})$$
$$\tau_r = \frac{\epsilon}{\sigma} \quad (\text{s}), \quad \rho_v(t) = \rho_{v0} e^{-t/\tau_r} \quad (\text{C/m}^3)$$

Cu: $\tau_r \approx 10^{-19}$ s; mica: $\tau_r \approx 10^4$ s.

DYNAMIC EM POTENTIALS

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$$

Retarded potentials *effects propagate at speed u_p , $R' = |\mathbf{R} - \mathbf{R}_i|$*

$$V(\mathbf{R}, t) = \frac{1}{4\pi\epsilon} \int \frac{\rho_v(\mathbf{R}_i, t - R'/u_p)}{R'} dV'$$
$$\mathbf{A}(\mathbf{R}, t) = \frac{\mu}{4\pi} \int \frac{\mathbf{J}(\mathbf{R}_i, t - R'/u_p)}{R'} dV'$$

TIME-HARMONIC POTENTIALS & PHASORS

$k = \omega/u_p$ (wavenumber); $e^{j\omega t}$ factor stripped to get phasor.

$$\tilde{V}(\mathbf{R}) = \frac{1}{4\pi\epsilon} \int \frac{\tilde{\rho}_v(\mathbf{R}_i)e^{-jkR'}}{R'} dV'$$
$$\tilde{\mathbf{H}} = \frac{1}{\mu} \nabla \times \tilde{\mathbf{A}}, \quad \tilde{\mathbf{A}} = \frac{\mu}{4\pi} \int \frac{\tilde{\mathbf{J}}(\mathbf{R}_i)e^{-jkR'}}{R'} dV'$$

Given $\tilde{\mathbf{E}}$: apply $\tilde{\mathbf{H}} = -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}}$ (Faraday in phasor domain).

LOSSLESS PLANE WAVE

Wave equation (phasor, source-free, $\sigma = 0$)

$$\nabla^2 \tilde{\mathbf{E}} + k^2 \tilde{\mathbf{E}} = 0, \quad k = \omega\sqrt{\mu\epsilon} \quad (\text{rad/m})$$

Intrinsic impedance

$$\eta = \sqrt{\frac{\mu}{\epsilon}} \quad (\Omega), \quad \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \approx 377 \, \Omega \approx 120\pi \, \Omega$$

Phase velocity & wavelength

$$u_p = \frac{1}{\sqrt{\mu\epsilon}} = \frac{\omega}{k} \quad (\text{m/s}), \quad \lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad (\text{m})$$

Fields (wave in $+\hat{z}$, $\mathbf{E}$ in $\hat{x}$)

$$\tilde{\mathbf{E}}(z) = E_{x0}^+ e^{-jkz} \hat{x}, \quad \tilde{\mathbf{H}}(z) = \frac{E_{x0}^+}{\eta} e^{-jkz} \hat{y}$$
$$\mathbf{E}(z, t) = |E_{x0}^+| \cos(\omega t - kz + \phi^+) \hat{x}$$
$$\mathbf{H}(z, t) = \frac{|E_{x0}^+|}{\eta} \cos(\omega t - kz + \phi^+) \hat{y}$$

TEM: $\tilde{\mathbf{E}} \perp \tilde{\mathbf{H}} \perp \hat{k}$. In lossless medium $\mathbf{E}$ and $\mathbf{H}$ are in *phase*.

DIRECTION RELATIONS

$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{k} \times \tilde{\mathbf{E}}, \quad \tilde{\mathbf{E}} = -\eta \hat{k} \times \tilde{\mathbf{H}}$$

$\hat{k}$	$\hat{E}$	$\hat{H} = \hat{k} \times \hat{E}$	Example $\tilde{\mathbf{E}}$
$+\hat{z}$	$+\hat{x}$	$+\hat{y}$	$E_0 e^{-jkz} \hat{x}$
$+\hat{z}$	$+\hat{y}$	$-\hat{x}$	$E_0 e^{-jkz} \hat{y}$
$-\hat{z}$	$+\hat{x}$	$-\hat{y}$	$E_0 e^{+jkz} \hat{x}$
$+\hat{x}$	$+\hat{y}$	$+\hat{z}$	$E_0 e^{-jkx} \hat{y}$

PHASOR $\leftrightarrow$ TIME DOMAIN

$$\cos(\omega t - kz) \longleftrightarrow e^{-jkz}$$
$$\cos(\omega t - kz + \phi) \longleftrightarrow e^{j\phi} e^{-jkz}$$
$$\sin(\omega t - kz) \longleftrightarrow -j e^{-jkz}$$
$$\partial/\partial t \longleftrightarrow j\omega \quad (\text{phasor domain})$$

Back to time: $\mathbf{E}(z, t) = \text{Re}\{\tilde{\mathbf{E}}(z)e^{j\omega t}\}$

FARADAY QUICK REF

$$\Phi = \iint \mathbf{B} \cdot d\mathbf{s} \quad (\text{Wb})$$
$$V_{\text{emf}} = -N d\Phi/dt$$
$$V^{\text{tr}} = -N \iint \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$
$$V^{\text{m}} = \oint (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

TRANSFORMER

$$V_1/V_2 = N_1/N_2$$
$$I_1/I_2 = N_2/N_1$$
$$Z_{\text{in}} = (N_1/N_2)^2 Z_L$$

Generator: $V_0 = N A \omega B_0$

MAXWELL PHASORS

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$$
$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}$$
$$\epsilon_c = \epsilon' - j\sigma/\omega$$
$$\tilde{\mathbf{J}}_d = j\omega\epsilon\tilde{\mathbf{E}}$$

WAVE PARAMS

$$k = \omega\sqrt{\mu\epsilon} \quad (\text{rad/m})$$
$$\eta = \sqrt{\mu/\epsilon} \quad (\Omega)$$
$$u_p = 1/\sqrt{\mu\epsilon} \quad (\text{m/s})$$
$$\lambda = 2\pi/k = u_p/f \quad (\text{m})$$

POTENTIALS & RELAX

$$\mathbf{B} = \nabla \times \mathbf{A}$$
$$\mathbf{E} = -\nabla V - \partial \mathbf{A}/\partial t$$
$$\tau_r = \epsilon/\sigma \quad (\text{s})$$
$$\rho_v = \rho_{v0} e^{-t/\tau_r}$$

POLARIZATION

*Polarization = curve traced by **E** in plane $\perp \hat{k}$ as t varies.*
General phasor (wave in $+\hat{z}$) $a_x = |E_{x0}|$, $a_y = |E_{y0}|$, δ = phase diff E_y rel. E_x

$$\tilde{\mathbf{E}}(z) = (a_x \hat{x} + a_y e^{j\delta} \hat{y}) e^{-jkz}$$
$$\mathbf{E}(z, t) = a_x \cos(\omega t - kz) \hat{x} + a_y \cos(\omega t - kz + \delta) \hat{y}$$

LINEAR POLARIZATION

Condition: $\delta = 0$ or $\delta = \pi$. E_x, E_y in/out of phase $\Rightarrow$ straight line.

$$\mathbf{E} = (a_x \hat{x} \pm a_y \hat{y}) \cos(\omega t - kz), \quad |\mathbf{E}| = \sqrt{a_x^2 + a_y^2} |\cos(\cdot)|$$
$$\psi = \tan^{-1} \left(\frac{\pm a_y}{a_x} \right) \text{ (tilt from } +\hat{x} \text{)}$$

$a_y = 0$: x -pol ($\psi = 0$); $a_x = 0$: y -pol ($\psi = \pm \pi/2$).

CIRCULAR POLARIZATION

Condition: $a_x = a_y = a > 0$, $\delta = \pm \pi/2$. $|\mathbf{E}| = a = \text{const.}$
LHC ($\delta = +\pi/2$)

$$\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}, \quad \mathbf{E} = a[\cos(\omega t - kz)\hat{x} - \sin(\omega t - kz)\hat{y}]$$

RHC ($\delta = -\pi/2$)

$$\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}, \quad \mathbf{E} = a[\cos(\omega t - kz)\hat{x} + \sin(\omega t - kz)\hat{y}]$$

ELLIPTICAL POLARIZATION

*General ($a_x \neq a_y$, $\delta \neq 0, \pm \pi/2, \pi$): **E** traces an ellipse.*

$$\psi_0 = \tan^{-1} \left(\frac{a_y}{a_x} \right) \in \left[0, \frac{\pi}{2} \right] \text{ (auxiliary angle)}$$
$$\tan 2\gamma = (\tan 2\psi_0) \cos \delta \quad \text{(rotation angle } \gamma \text{)}$$
$$\sin 2\chi = (\sin 2\psi_0) \sin \delta \quad \text{(ellipticity angle } \chi \text{)}$$
$$R = \frac{a_\xi}{a_\eta} \text{ (axial ratio),} \quad \tan \chi = \pm \frac{1}{R}$$

$\tan \chi > 0$: LH ; $\tan \chi < 0$: RH ; $R = 1$: *circular*; $a_\eta = 0$: *linear*.

EM WAVES IN LOSSY MEDIUM

Complex permittivity

$$\varepsilon_c = \varepsilon' - j\varepsilon'' = \varepsilon_r \varepsilon_0 - j \frac{\sigma}{\omega}, \quad \frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega \varepsilon'}$$

Propagation constant $\gamma = \alpha + j\beta$; $\gamma^2 = -\omega^2 \mu \varepsilon_c$

$$\alpha = \omega \sqrt{\frac{\mu \varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} - 1 \right]^{\frac{1}{2}}} \text{ (Np/m)}$$
$$\beta = \omega \sqrt{\frac{\mu \varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} + 1 \right]^{\frac{1}{2}}} \text{ (rad/m)}$$

Intrinsic impedance $\eta_c = |\eta_c| e^{j\theta_\eta}$, *complex*

$$\eta_c = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon'}} \left(1 - j \frac{\varepsilon''}{\varepsilon'} \right)^{-1/2}$$

Fields (wave in $+\hat{z}$, $\mathbf{E} \parallel \hat{x}$)

$$\tilde{\mathbf{E}}(z) = E_{x0} e^{-\alpha z} e^{-j\beta z} \hat{x}, \quad \tilde{\mathbf{H}}(z) = \frac{E_{x0}}{\eta_c} e^{-\alpha z} e^{-j\beta z} \hat{y}$$

E, H not in phase in lossy media (phase diff θ_η).

MEDIUM CLASSIFICATION

Type	$\varepsilon''/\varepsilon'$	Approximations
Low-loss diel.	$\ll 1$	$\alpha \approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\varepsilon'}}, \beta \approx \omega \sqrt{\mu \varepsilon'}, \eta_c \approx \sqrt{\mu/\varepsilon'}$
Quasi-cond.	$\frac{10^{-2}}{10^2}$	– Use general α, β
Good cond.	$\gg 100$	$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}, \eta_c \approx (1 + j)\sqrt{\pi f \mu / \sigma}$

Skin depth $|\mathbf{E}|$ drops to 1/e of surface value

$$\delta_s = \frac{1}{\alpha} \text{ (m)}, \quad \delta_s^{\text{cond}} = \frac{1}{\sqrt{\pi f \mu \sigma}} \text{ (m)}$$
$$|\tilde{\mathbf{E}}(z)| = |E_{x0}| e^{-\alpha z}; \quad |\tilde{\mathbf{H}}(z)| = \frac{|E_{x0}|}{|\eta_c|} e^{-\alpha z}.$$

CURRENT IN GOOD CONDUCTOR

AC current flows near surface; DC current uniform throughout conductor. Current density at depth z $J_0 = \sigma E_0$, E_0 at surface

$$\tilde{\mathbf{J}}(z) = J_0 e^{-(1+j)z/\delta_s} \hat{x} \text{ (A/m}^2\text{)}$$

Total current, slab width w (y -dir)

$$\tilde{I} = \frac{J_0 w \delta_s}{1 + j} \text{ (A)}$$

Surface impedance

$$Z_s = \frac{1 + j}{\sigma \delta_s} = R_s(1 + j), \quad R_s = \sqrt{\frac{\pi f \mu}{\sigma}} \text{ (}\Omega\text{)}$$

AC resistance: slab (width w , length ℓ)

$$R = R_s \frac{\ell}{w} \text{ (}\Omega\text{)}, \quad R'_{\text{coax}} = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) \text{ (}\Omega/\text{m)}$$

EM POWER DENSITY (POYNTING)

Instantaneous Poynting vector W/m^2

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \text{ (W/m}^2\text{)}$$

Average power density (time-harmonic)

$$\mathbf{S}_{\text{av}} = \frac{1}{2} \text{Re} \{ \tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \} \text{ (W/m}^2\text{)}$$

Lossless ($\alpha = 0$, $\theta_\eta = 0$, η real)

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{k} \text{ (W/m}^2\text{)}$$

Lossy ($\eta_c = |\eta_c| e^{j\theta_\eta}$, wave in $+\hat{z}$)

$$\mathbf{S}_{\text{av}} = \frac{|\tilde{E}(0)|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \hat{z} \text{ (W/m}^2\text{)}$$

Power $\propto e^{-2\alpha z}$, field $\propto e^{-\alpha z}$: power decays at 2 $\times$ the rate.
Power through area A , angle θ to $\hat{n}$

$$P = S_{\text{av}} A \cos \theta \text{ (W)}$$

ATTENUATION & DECIBELS

Unit conversion

$$\alpha \text{ (dB/m)} = 8.69 \alpha \text{ (Np/m)}$$

Power attenuation rate from $z = 0$

$$A = 10 \log \left(\frac{S_{\text{av}}(z)}{S_{\text{av}}(0)} \right) = -8.69 \alpha z \text{ (dB)}$$

Field attenuation from $z = 0$

$$A = 20 \log \left(\frac{|E(z)|}{|E(0)|} \right) \text{ (dB)}$$

Power/voltage ratio

$$G = 10 \log \left(\frac{P_1}{P_2} \right) = 20 \log \left(\frac{V_1}{V_2} \right) \text{ (dB)}$$

$P \times 2 \Rightarrow +3 \text{ dB}; \quad P \times 10 \Rightarrow +10 \text{ dB}; \quad |E| \times 2 \Rightarrow +6 \text{ dB.}$

READ-A-WAVE WORKFLOW

Given $\mathbf{E}(z, t) = E_0 \cos(\omega t - kz + \phi) \hat{a}$:

- Read: ω, k, E_0, ϕ , dir. $\hat{a}$, prop. dir. $\hat{k}$
- $f = \omega/2\pi$; $\lambda = 2\pi/k$; $u_p = \omega/k$
- $\eta = \sqrt{\mu/\varepsilon}$ (or μu_p); free space: $\eta_0 \approx 377 \Omega$
- Phasor: $\tilde{\mathbf{E}} = E_0 e^{j\phi} e^{-jkz} \hat{a}$
- $\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{k} \times \tilde{\mathbf{E}}$; $\mathbf{H}(z, t) = \text{Re}\{\tilde{\mathbf{H}} e^{j\omega t}\}$
- $S_{\text{av}} = |E_0|^2/(2\eta) \text{ (W/m}^2\text{)}$

POLARIZATION TYPES

Linear: $\delta = 0$ or π
LHC: $a_x = a_y$, $\delta = +\pi/2$
RHC: $a_x = a_y$, $\delta = -\pi/2$
Ellip.: general a_x, a_y, δ

LOSSY PARAMS

$$\varepsilon_c = \varepsilon' - j\sigma/\omega$$

LHC: $a_x = a_y$, $\delta = +\pi/2$
RHC: $a_x = a_y$, $\delta = -\pi/2$
Ellip.: general a_x, a_y, δ

MEDIUM TYPES

$$\varepsilon''/\varepsilon' \ll 1$$
: low-loss diel.
 $10^{-2} \leq \varepsilon''/\varepsilon' \leq 10^2$: quasi
 $\varepsilon''/\varepsilon' \gg 100$: good cond.
 $\varepsilon'' = \sigma/\omega$, $\varepsilon' = \varepsilon_r \varepsilon_0$

POWER DENSITY

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \text{ (inst.)}$$
$$S_{\text{av}} = |\mathbf{E}_0|^2/(2\eta) \text{ (lossless)}$$
$$S_{\text{av}} = \frac{|\mathbf{E}(0)|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta$$
$$P = S_{\text{av}} A \cos \theta$$

dB QUICK REF

$$\alpha_{\text{dB}} = 8.69 \alpha_{\text{Np}}$$
$$G = 10 \log(P_1/P_2)$$
$$A = -8.69 \alpha z \text{ (dB)}$$
$$A = 20 \log(|E(z)|/|E(0)|)$$